ELECTRIC SECTOR CONTROL-COMPANY FUNCTIONAL CLASS

Functional Capability Design Basis: Physical, Human, and Software-Data Layers

DBA-ES-CntlCo: Balancing Authority, Reliability Coordinator, and Transmission Operator

A Functional Class Instrument Under

DBA-ES: Electric Sector Design Basis Framework (v0.6)

DBA-ES FC5 and CntlCo Gap Analysis (v0.1) as Predecessor Document

Version 0.1 — DRAFT

DBA Consequence Envelope and MA Extension Deferred to v0.2

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For review by ISO/RTO operations staff, NERC reliability analysts, balancing authority engineers, FERC staff, and policy officials responsible for bulk electric system reliability governance.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. Functional capability design basis (DB layer only); DBA consequence envelope and MA extension deferred to v0.2. Primary scope: Balancing Authority (BA) function as governing case; Reliability Coordinator (RC) and Transmission Operator (TOP) addressed as functional extensions. Three-layer design basis structure established: Physical Layer (PL), Human Layer (HL), and Software-Data Layer (SL). Cross-sector cascade category (XC) retained from DBA-ES sector parent. Four locally-defined event categories are instantiated with CntlCo-specific events. Established two-way dependency relationship with FC instruments: CntlCo functional degradation produces FC-level consequences; FC-level events produce CntlCo data feed degradation. Bright Line and Command Broker architecture applied to CntlCo AI operational tools. Acknowledged dual-DB architecture per DBA-ES §3.4.

Section 1 — Purpose, Instrument Type, and Version Status

1.1 Purpose and Instrument Type

This document establishes the functional capability Design Basis for the electric sector control-company function within the Electric Sector Design Basis Series. It is a functional class instrument — not a physical facility-class design basis — and is the only instrument of this type in the DBA-ES series. The distinction matters analytically: where a facility-class instrument asks what happens if a building or piece of equipment is damaged, this instrument asks what happens if the CntlCo's operational function is degraded or denied.

The CntlCo function — the balancing authority, reliability coordinator, and transmission operator functions as defined by NERC functional registration — is a licensed operational capability: a set of responsibilities and authorities supported by technology and executed by certified personnel. It is cross-cutting: it governs GenCo, TransCo, and DisCo entities simultaneously. Its degradation produces consequences across all facility classes in the DBA-ES series. Every FC instrument's MA extension must address the consequences of CntlCo degradation for that facility class; this instrument establishes the CntlCo's own design basis that those FC-level analyses draw upon.

This is a DB-layer instrument. The DBA consequence envelope and the military action extension are deferred to v0.2. The three-layer design basis structure — physical, human, and software-data — is established here as the analytical foundation for those subsequent layers.

1.2 NERC Functional Registration Scope

The CntlCo instrument addresses three NERC-registered functions. The Balancing Authority (BA) function is the primary case: it is the function most directly responsible for real-time generation-load balance. It is the natural governing unit for the design basis. The Reliability Coordinator (RC) and Transmission Operator (TOP) functions are addressed as extensions in Section 5.

NERC Function	Primary Responsibility	Geographic Scope	FC Instrument Dependency Relationship
Balancing Authority (BA)	Maintains real-time balance between generation and load within the BA area. Controls AGC, manages reserves, and coordinates with neighboring BAs.	BA area: typically one or more control areas within a single interconnection. May span multiple states.	Directly dispatches GenCo facilities (FC4 gas-fired generation via gas-electric cascade awareness). Receives TransCo facility status from all FC instruments. Primary CntlCo function for this instrument.
Reliability Coordinator (RC)	Maintains reliability of the bulk electric system across a wide area. Has the highest level of authority during system emergencies. Oversees multiple BA areas.	RC area: typically broader than a single BA area; may span multiple BAs and states. In North America, RC areas cover large portions of each interconnection.	Oversees all FC instrument consequence events within its area. Issue emergency directives to BA, TOP, and GO functions. RC functional degradation removes the highest authority layer from emergency response.
Transmission Operator (TOP)	Operates transmission facilities within its TOP area in real time. Implements directives from the RC and BA. Controls switching operations on transmission equipment.	TOP area: typically corresponds to a single Transmission Owner's footprint. Smaller geographic scope than BA or RC.	Directly operates FC2 (EHV substations) and FC3 (HVDC interconnects) facilities. TOP functional degradation removes local switching authority for transmission infrastructure.

1.3 Predecessor Document

The DBA-ES FC5 and CntlCo Gap Analysis (v0.1, June 2026) established the CntlCo's architectural rationale as a functional class instrument, distinguished it from the physical facility classes, and identified the three-layer design basis structure. This instrument develops the gap analysis architecture into a full DB-layer instrument. Where this document differs from the gap analysis, this document governs.

1.4 Position in Series

Document Number: DBA-ES-CntlCo-001

Series: Electric Sector Design Basis Series — Functional Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.6)

Predecessor Document: DBA-ES FC5 and CntlCo Gap Analysis (v0.1)

Relationship to FC instruments: two-way dependency. CntlCo functional degradation produces consequences at each FC instrument facility class. FC instrument CE events (data feed loss, SCADA disruption) produce SL-layer events in this instrument. Neither document substitutes for the other.

Section 2 — The Three-Layer Design Basis Structure

2.1 Rationale for the Three-Layer Framework

The three-layer framework — physical, human, and software-data — is the organizing principle for the CntlCo functional capability design basis because the CntlCo's operational capability depends simultaneously on all three layers and can be degraded by failure of any one of them, even when the other two remain fully functional. This is the defining characteristic that distinguishes the CntlCo functional class from the physical facility classes: a substation can carry load passively without human operators or sophisticated software; the CntlCo function cannot exist without all three layers operating together.

The three-layer framework also reflects how the CntlCo function has failed historically. The 2003 Northeast Blackout was primarily a software-data layer failure: the operators were present, and the physical infrastructure was largely intact, but the EMS alarm system failure and state estimator degradation removed operator situational awareness. Future failures under military action conditions are more likely to target the human layer directly — removing qualified operators through access denial, cognitive overload, or incapacitation — because the physical and software layers have received more security investment than the human layer.

2.2 Layer Definitions and Interdependencies

Layer	Components	Failure Mode	Consequence if Degraded
Physical Layer (PL)	Primary control center building and infrastructure. Backup control center (BCC). Dedicated communications infrastructure (microwave, fiber, satellite backup). Computing hardware	Physical damage to the control center or BCC. Communications infrastructure failure. Computing hardware failure. Loss of control center power supply.	Loss of the physical node that hosts the CntlCo function. If the primary control center is lost and the BCC transition fails, the CntlCo function is unavailable.

	(EMS servers, workstations, networking).		
Human Layer (HL)	Licensed system operators holding NERC system operator certifications. Shift supervisors. Operations management. Training and qualification infrastructure.	Operator incapacitation. Staffing below the minimum crew complement. Training currency lapse. Loss of qualified operators during an extended emergency.	Degraded operational decision quality. Reduced ability to respond to complex simultaneous contingencies. Under military action: deliberate targeting of operator staff.
Software-Data Layer (SL)	Energy Management System (EMS). State estimator. Contingency analysis tools—Automatic Generation Control (AGC). SCADA data feeds from GenCo, TransCo, and DisCo facilities—real-time and historical databases.	EMS software failure. State estimator divergence from bad data inputs. SCADA feed loss from FC instrument facilities—cyber intrusion modifying EMS data or algorithms.	The operator loses reliable situational awareness—decisions are made on incorrect system state. Contingency analysis produces incorrect results. AGC operates on incorrect frequency or interchange data.

The three layers are not independent. They interact at specific interfaces that are themselves design basis concerns. The physical layer hosts the software-data layer: EMS servers require power, cooling, and network connectivity from the physical layer infrastructure. The software-data layer informs the human layer: operators make decisions based on the situational picture the EMS provides. The human layer activates the physical layer's emergency capabilities: BCC transitions, backup communications activation, and manual operations mode all require human initiation. A compound failure that simultaneously degrades two layers — physical damage that also disrupts data feeds (PL-4 producing SL-4), or cyber intrusion that both corrupts EMS data and removes operator confidence in the system picture (SL-5 producing HL-2) — is the governing design basis concern for the CntlCo.

2.3 Regulatory Context

NERC functional registration requirements establish the legal framework for BA, RC, and TOP functions. Entities holding these registrations are subject to NERC reliability standards enforcement and FERC oversight. The CIP standards series (CIP-002 through CIP-014) establishes cybersecurity requirements for critical cyber assets associated with the reliable operation of the bulk electric system, including EMS systems and control center infrastructure. CIP-014 Physical Security establishes requirements for transmission stations and substations, but does not directly address control center physical security to the same standard.

No existing NERC standard requires a BA to maintain a functional capability design basis — a defined set of postulated degradation conditions that the BA function must be demonstrated to withstand. NERC standards require specific capabilities (BCC, EMS redundancy, operator certification) without framing them as a coherent design basis. This instrument supplies that framing.

FERC Order 693 and the associated reliability standards establish planning and operational requirements that the CntlCo function implements. The design basis established here is consistent with and complementary to those requirements, addressing the consequence analysis that the standards framework implies but does not explicitly require.

Section 3 — Physical Layer (PL) Postulated Events

Physical layer events address the infrastructure that hosts the CntlCo function: the control center buildings, backup control center, communications infrastructure, and computing hardware. PL events do not directly degrade the CntlCo's analytical capability or operator quality; they degrade or remove the physical platform on which those capabilities run. The governing PL concern is not any single event but the common-mode failure between the primary control center and BCC that removes both hosting platforms simultaneously.

Code	Event	Initiating Condition	Governing Concern
PL-1	Primary control center loss — power	Loss of AC power supply to the primary control center building, removing power from EMS servers, operator workstations, display systems, communications equipment, and HVAC. UPS provides bridge power; the backup generator provides extended power if the utility supply is not restored.	Primary control center power loss is the analog of station service loss in the FC instrument series. The CntlCo function continues under UPS and generator power, but the UPS design duration and generator fuel endurance bound the available response window. If neither restores within the design duration, the BCC transition must be executed.
PL-2	Primary control center loss — physical damage	Physical damage to the primary control center building from fire, flood, structural failure, or external physical event, rendering the control center uninhabitable or inoperable. CntlCo function must be transferred to the BCC.	BCC transition is the governing design basis response to PL-2. The transition must be executable within the design basis transition time — the period during which the CntlCo function can be maintained at degraded capability from the damaged primary control center before the BCC must be operational. BCC transition time and the interim degraded capability are the primary acceptance criteria.
PL-3	Backup control center (BCC) unavailability	The BCC is unavailable at the time it is needed: it has not been maintained to operational readiness, it is affected by the same event that damaged the primary	PL-3 is the compounding event that transforms PL-2 from a managed contingency into a CntlCo functional loss. The

		control center, or it is simultaneously under its own maintenance outage. The CntlCo must sustain function without an available BCC.	design basis must require geographic and infrastructure independence between the primary control center and BCC such that a single event cannot disable both simultaneously. Common-mode failure between primary and BCC is the governing design basis concern.
PL-4	Communications infrastructure failure	Loss or severe degradation of the communications infrastructure connecting the control center to the FC instrument facilities it monitors and controls: SCADA polling failure, loss of voice communications with field operators, loss of inter-BA communications circuits, or loss of the RC-to-BA authority chain communications.	PL-4 simultaneously degrades the SL layer (loss of real-time data feeds) and the HL layer (loss of operator-to-field communications). The CntlCo's ability to manage the bulk electric system degrades in proportion to the completeness of the communications loss. Partial loss produces degraded situational awareness; complete loss produces operational blindness.
PL-5	BCC transition failure — incomplete or delayed	A BCC transition that is initiated following PL-1 or PL-2 but does not complete successfully: the BCC EMS does not synchronize to the current system state, operator connectivity is not established, or real-time data feeds are not restored at the BCC. The CntlCo function is in an indeterminate state — neither the primary control center nor the BCC is fully operational.	PL-5 is the mode transition analog for the CntlCo: the failed transition to BCC is structurally similar to MT-4 (failed transition) in DBA-ES-DI-FC5. The design basis must specify the maximum acceptable BCC transition time and the minimum BCC operational capability that constitutes a successful transition.

Section 4 — Human Layer (HL) Postulated Events

Human layer events address the certified system operators, their training currency, their physical availability, and the authority chain through which they receive and execute emergency directives. HL events are the least institutionally addressed of the three layers: NERC certification establishes minimum qualification standards, but the design basis for operator availability, cognitive capacity, and authority chain integrity under degraded conditions is not systematically addressed in current reliability standards.

The deliberate targeting of the human layer — through access denial (HL-4), cognitive overload from simultaneous contingency generation (HL-2), or disruption of the authority chain (HL-5) — is an MA-layer concern addressed in the extension deferred to v0.2. At the DB layer, HL events are postulated without adversarial intent; the same failure modes can result from natural causes, staffing shortfalls, or infrastructure failures.

Code	Event	Initiating Condition	Governing Concern
HL-1	Staffing below the minimum qualified operator complement	The on-duty crew falls below the minimum number of certified system operators required to maintain all required CntlCo functions simultaneously. May result from illness, injury, failure to report, or an extended emergency that exhausts crew rotation capability.	NERC system operator certification requirements establish minimum qualification standards but do not specify minimum crew complement requirements for all operating conditions. The design basis must define the minimum qualified crew complements for each operating mode and the consequences of falling below them.
HL-2	Operator cognitive degradation under sustained emergency	An extended duration emergency requiring continuous operator engagement beyond normal shift rotation capability, producing fatigue-induced cognitive degradation. Simultaneous multi-contingency events — the type most likely under coordinated military action — impose the highest cognitive load and are most likely to exceed individual operator capacity.	Human performance degradation under cognitive overload is a documented failure mode in grid operations. The 2003 Northeast Blackout investigation identified operator cognitive overload as a contributing factor. Under military action conditions, an adversary who understands the relationship between simultaneous contingency count and operator cognitive capacity can deliberately exceed that capacity.
HL-3	Training currency lapse for emergency procedures	Certified system operators who have not exercised emergency procedures — BCC transition, manual load shedding, black start coordination — within the required training currency window. Emergency procedures not exercised regularly produce incorrect or incomplete execution when required in real conditions.	Training currency is a design basis requirement, not a procedural aspiration. An emergency procedure that has not been exercised within the design basis currency window is not available as a design basis response. The BA’s design basis must account for the possibility that trained operators are not available for specific procedures when needed.
HL-4	Loss of qualified operator access to the control center	Qualified operators cannot reach the primary control center or BCC to assume or relieve shifts. May result from access denial conditions (road closures, security perimeter lockdown), incapacitation of operators en route, or denial of physical access to the control center facility.	HL-4 is the human layer analog of physical denial in the MA framework. Under military action conditions, preventing qualified operators from reaching the control center is a viable adversarial strategy that does not require any physical attack on the control center itself. The design basis must address remote operations capability and extended shift endurance as responses to access denial.

HL-5	Command authority chain degradation	Degradation or disruption of the authority chain from RC to BA to TOP that governs emergency operations. May result from loss of RC communications (PL-4), RC functional unavailability (RC extension event), or deliberate disruption of inter-entity coordination during an emergency.	The RC holds the highest authority in bulk electric system emergencies. If RC communications are lost, the BA must operate under its own authority within defined protocols. The design basis must address the BA's operating authority and decision-making framework under conditions of RC communications loss.

Section 5 — Software-Data Layer (SL) Postulated Events

Software-data layer events address the EMS, state estimator, AGC, contingency analysis tools, and the real-time data feeds from FC instrument facilities that give those systems their situational picture. The SL layer is the CntlCo's analytical core: degradation of the SL layer does not remove the CntlCo's authority or physical platform, but it removes the operator's ability to exercise that authority correctly.

SL-1 (state estimator divergence) is identified as the governing SL event because all downstream EMS functions depend on a correct state estimator solution. An incorrect state estimator solution that is not flagged as diverged is more dangerous than an acknowledged EMS failure: it produces operator decisions that are confident but wrong. The adversarial bad data injection pathway to SL-1 — deliberately manipulated SCADA data from compromised FC instrument facilities — is a precision attack on the SL layer that is analytically distinct from the random measurement error case and must be treated as a separate design basis concern.

Code	Event	Initiating Condition	Governing Concern
SL-1	State estimator divergence from bad data	The EMS state estimator produces an incorrect system state solution due to bad data inputs: failed sensors, communications errors, or deliberately manipulated SCADA measurements from FC instrument facilities. The state estimator may converge to an incorrect solution without flagging the divergence, presenting operators with a system picture that does not reflect actual conditions.	State estimator divergence is the most consequential SL event because it corrupts all downstream functions that depend on state estimation: contingency analysis, security assessment, AGC base points, and interchange scheduling. An operator making decisions on an incorrect state estimator solution is making decisions on a false picture of the system. Under military action conditions, deliberate injection of false SCADA data into FC instrument feeds is a precision attack on the SL layer.

SL-2	EMS software failure or corruption	Failure of the EMS application software through software defect, hardware failure supporting the EMS platform, or deliberate corruption through cyber intrusion. May affect a single EMS application (state estimator, contingency analysis) or the entire EMS platform.	EMS software failure without a functioning backup. EMS forces the CntlCo to operate using manual calculations, historical data, and voice communications. This degraded mode was the standard before modern EMS, but current operators may not be trained or equipped to sustain it. The transition to manual operations mode is an HL event as much as an SL event.
SL-3	AGC failure or incorrect operation	Automatic Generation Control failure, producing uncontrolled frequency or interchange deviation within the BA area. AGC failure may result from software failure, communications loss to generation units, or incorrect setpoints produced by a corrupted state estimator feeding AGC base points.	AGC failure in a large BA area can produce frequency deviations that trigger automatic protective responses — generator governor response, UFLS relay operation — before operators can manually intervene. The cascade from AGC failure to UFLS operation is a design basis concern that the 2003 Northeast Blackout investigation documented as a contributing pathway.
SL-4	Loss of real-time data feeds from FC instrument facilities	Loss of SCADA data feeds from one or more FC instrument facility classes: GenCo generation status, TransCo transmission facility status (FC2, FC3), DisCo distribution substation status (FC5), or compressor station pipeline status (FC4). The CntlCo loses real-time visibility into the operating state of the facilities it manages.	SL-4 establishes the two-way dependency between the CntlCo instrument and the FC instruments. FC instrument CE events that remove SCADA visibility from the CntlCo (CE-1 events in FC2, FC3, FC4, FC5) are simultaneously SL-4 events at the CntlCo level. The compound loss of multiple FC instrument data feeds — from a coordinated attack targeting multiple facilities' communications simultaneously — produces a state estimator divergence (SL-1) of greater severity than any single feed loss.
SL-5	Cyber intrusion into the EMS or SCADA infrastructure	Unauthorized access to the EMS platform, SCADA systems, or the communications infrastructure connecting the CntlCo to FC instrument facilities. Active manipulation can corrupt state estimator inputs (leading to SL-1), modify AGC setpoints (leading to SL-3), issue unauthorized control commands to field equipment, or exfiltrate system topology and operating data.	CntlCo cyber intrusion is a high-value adversarial objective because the CntlCo has command authority over the entire BA area's generation and transmission infrastructure. A compromised CntlCo is not merely blind — it may issue incorrect or harmful commands to FC instrument facilities across the entire BA area simultaneously. NERC CIP standards address

			CntlCo cybersecurity but do not establish a design basis for the consequences of a successful intrusion.

Section 6 — Cross-Sector Cascade (XC) Postulated Events

Cross-sector cascade events from the CntlCo instrument operate at the largest geographic scale in the DBA-ES series. Where FC2 cascade events affect a metropolitan load pocket, and FC3 cascade events affect a regional interconnection, CntlCo cascade events affect the entire BA area — and, through the RC extension scope, potentially an entire interconnection sub-region.

The XC events for the CntlCo are also distinctive in their mechanism. Unlike the FC instrument XC events, which cascade outward from a physical facility loss, the CntlCo XC events cascade from a governance and awareness failure. The system may be physically intact; it is the CntlCo's inability to manage it correctly that produces the cascade. This makes the CntlCo XC events more insidious than FC-level physical cascade events: they may not be immediately visible as a cascade initiator, and they may produce consequences that are attributed to operator error rather than to the underlying design basis failure.

Code	Event	Initiating Condition	Governing Concern
XC-1	Bulk electric system reliability degradation from CntlCo functional loss	Degradation or loss of the CntlCo function produces degraded management of the bulk electric system within the BA area. Generation dispatch becomes suboptimal, transmission constraints go unmanaged, and the system operates with reduced awareness of its proximity to reliability limits. Subsequent contingency events produce larger consequences than they would under normal CntlCo function.	This is the primary XC event for the CntlCo: the BA's functional degradation does not immediately cause a blackout but reduces the system's ability to withstand subsequent events. The 2003 Northeast Blackout established this pattern definitively — operator awareness loss preceded and enabled the cascade. XC-1 is the design basis translation of that lesson.
XC-2	Cascade initiation from unmanaged constraint violation	Under degraded CntlCo function, a transmission constraint is violated and goes undetected or unresponded to. The constraint violation triggers an automatic protection response — line trips, generator trips, or UFLS — that initiates a cascade. The cascade propagates in proportion to the severity of the constraint violation and the pre-existing stress on the system.	XC-2 is the mechanism by which SL-1 (state estimator divergence) or SL-4 (data feed loss) produces a Tier 2 consequence. The CntlCo did not dispatch generation incorrectly; it managed the system correctly, given incorrect data. The consequence is indistinguishable from an operator error, but the root cause is a data integrity failure in the SL layer.

XC-3	Cross-sector consequence through load shedding	Load shedding initiated under CntlCo functional degradation — either as an emergency response to XC-2 cascade or as a managed load reduction under degraded situational awareness — removes power from critical facilities in the BA area. The Tier 3 pathway runs from CntlCo functional degradation through bulk electric system reliability loss through load shedding to critical service interruption.	XC-3 establishes that the CntlCo's design basis must address the Tier 3 consequence of its own functional degradation. No current NERC standard requires a BA to demonstrate that its operational continuity measures are adequate to prevent XC-3 consequences. This design basis gap is analogous to the gaps identified in the FC instrument series for long-lead-time equipment and extended consequence regimes.
XC-4	RC-area consequence from coordinated multi-BA CntlCo attack	Simultaneous or sequential degradation of multiple BA CntlCo functions within a single RC area, removing the RC's ability to manage the area as a coordinated reliability zone. The RC cannot issue effective emergency directives if the BA functions it oversees are simultaneously degraded.	XC-4 operates at the RC-area geographic scale. It is the CntlCo-level analog of the coordinated multi-node attack in DBA-ES-TX-FC2: targeting multiple BA control centers simultaneously defeats the redundancy that the RC-to-BA hierarchy provides. The RC extension scope is the primary analytical context for XC-4.

Section 7 — RC and TOP Functional Extensions

7.1 Extension Rationale

The RC and TOP functions are addressed as extensions to the BA primary case rather than as separate instruments because they share the same three-layer design basis framework and the same four postulated event categories. The RC and TOP extensions differ from the BA primary case in geographic scope, authority level, and the specific FC instruments whose facilities they operate. The following table identifies the additional design basis elements and open items specific to each extension.

Function	Additional Design Basis Elements Beyond BA	Open Items Specific to This Extension
Reliability Coordinator (RC)	Geographic scope: RC area spans multiple BA areas, requiring the RC's design basis to address simultaneous degradation of multiple BA CntlCo functions within the RC area (XC-4). Authority chain: RC holds the highest emergency authority; RC functional loss removes the governing authority layer	RC-specific BCC requirements: the RC's BCC must maintain wide-area situational awareness from a geographically independent location. RC minimum operator complement: the RC's operator complement must address the broader geographic scope. RC-to-BA authority chain communications

	from the entire RC area simultaneously. Wide-area situational awareness: RC's SL layer must integrate data from all BA areas within the RC area; the data volume and diversity are substantially greater than a single BA's SL layer.	design basis: the mechanism for maintaining RC authority during primary control center loss is not addressed in this v0.1 instrument and is deferred to v0.2.
Transmission Operator (TOP)	Physical proximity to FC instruments: the TOP's operational scope directly includes FC2 (EHV substations) and FC3 (HVDC interconnects) facilities; TOP functional degradation removes the switching authority for those facilities. Local scope: the TOP's geographic scope is smaller than the BA or RC, making its design basis more closely tied to the physical FC instruments in its footprint. TOP operators perform physical switching operations at the substation level and must be able to do so under conditions of CntlCo PL and HL degradation.	TOP field operations under CntlCo degradation: the TOP's ability to perform manual switching at FC2 and FC3 facilities when the SCADA path is unavailable (PL-4, SL-4) requires specific procedures and training that are not addressed in v0.1. The TOP minimum crew complement for simultaneous FC-level events is deferred to v0.2.

7.2 RC Extension: Governing Concern

The RC's governing design basis concern is XC-4: coordinated degradation of multiple BA CntlCo functions within the RC area. If the RC's own CntlCo function is simultaneously degraded — through a direct attack on the RC control center or through PL-4 communications loss, severing the RC from its BA area — the RC area loses its governing reliability authority at the same moment that multiple BA functions are degraded. This compound scenario is the RC-level analog of the coordinated multi-node attack in the FC instrument series.

The RC extension design basis must address this compound scenario explicitly and must establish that the RC's BCC and authority chain communications are independent of the BA control centers within the RC area. A common physical infrastructure connecting the RC and its BA control centers — a shared communications backbone, a collocated BCC — would create a common-mode vulnerability across the entire RC area that is analytically equivalent to PL-3 (BCC unavailability) at the RC level.

7.3 TOP Extension: Governing Concern

The TOP's governing design basis concern is the loss of switching authority over FC2 (EHV substations) and FC3 (HVDC interconnects) facilities under conditions of CntlCo PL or SL degradation. When the TOP's SCADA path to those facilities is unavailable (PL-4 producing SL-4), the TOP must execute switching operations through field personnel using voice communications and manual procedures. The TOP's design basis must address this manual operations mode for field switching as a required capability — not a fallback of last resort —

because it is the governing response to the most likely CntlCo degradation scenario under military action.

Section 8 — DB-Layer Acceptance Criteria

The following table establishes the DB-layer acceptance criteria for the CntlCo functional class. DBA-level acceptance criteria will be established in v0.2.

Criterion	Requirement	Source / Rationale
BCC geographic independence	The primary control center and BCC must be geographically and infrastructure-independent such that no single physical event — natural hazard, precision strike, or area denial — can simultaneously disable both. Geographic separation must be sufficient to place the BCC outside the design basis physical damage radius and outside the design basis natural hazard zone.	Derived from PL-3 as the governing concern. Common-mode failure between primary and BCC is the failure mode that transforms a managed contingency into CntlCo functional loss.
BCC transition time	The BCC must be capable of assuming full CntlCo function within a design basis transition time to be specified in the site-specific implementation. During the transition period, the degraded primary control center must maintain a minimum CntlCo function. The minimum CntlCo function during transition must include: AGC continuity, RC communications, and emergency load shedding authority.	Derived from PL-2 and PL-5. The transition time design basis must be calibrated against the time required for an unmanaged BA area to reach a constraint violation that initiates cascade (XC-2)— deferred quantification: DI-1.
Minimum qualified operator complement	The BA must maintain a minimum qualified operator complement on each shift sufficient to perform all required real-time functions simultaneously: frequency control, interchange scheduling, emergency load shedding, and RC communications. The minimum complement must be specified and must account for the cognitive load of simultaneous multi-contingency events.	Derived from HL-1 and HL-2. Minimum complement specification is deferred to site-specific implementation; the design basis requirement to specify it is established here.
State estimator data integrity	The EMS state estimator must include bad data detection and rejection capability sufficient to prevent SL-1 divergence from corrupted SCADA inputs. The bad data detection must be effective against both random measurement errors and systematically manipulated data from adversarially compromised FC instrument facilities.	Derived from SL-1 as the governing SL concern. Adversarial bad data injection is a precision SL-layer attack vector; the design basis must treat it explicitly, not only as a random measurement error problem.

Independent CntlCo communications path	The CntlCo must maintain at least one communications path to each FC instrument facility that is physically independent of the primary SCADA communications infrastructure. This independent path must support emergency voice coordination when SCADA is unavailable and must be survivable under the PL-4 design basis communications failure scenario.	Derived from PL-4 and the Command Broker communications independence requirement established in DBA-ES-DI-FC5 §3.3. The CntlCo is itself a Command Broker authority for FC instrument facilities; its communications independence requirement follows the same logic.
Manual operations mode capability	The CntlCo must maintain the capability to operate the BA area in manual mode — without functional EMS, state estimator, or AGC — for a minimum design basis duration using manual calculations, voice communications, and historical data. Operators must be trained, and the manual operations procedures must be current within the design basis training currency window.	Derived from SL-2 and HL-3. Manual operations capability is the fallback for the complete SL layer failure scenario. Without it, the CntlCo's design basis response to SL-2 is functionally empty.
Bright Line for CntlCo AI tools	ML systems used in state estimation, contingency analysis, and AGC optimization operate below the Bright Line and may operate autonomously. Gen AI systems used in operational decision support, emergency procedure guidance, or situational awareness synthesis require a human Command Broker above the Bright Line. The Command Broker for CntlCo Gen AI tools is the licensed system operator on duty; this function cannot be delegated to a non-licensed person.	Derived from DBA-ES §3.4 and DBA-DC-L1 §3.5. The licensed system operator is the natural Command Broker for CntlCo AI tools: NERC certification establishes the qualification standard; the operator's legal accountability under NERC's rules of procedure provides the accountability framework.

Section 9 — Dual Design Basis Architecture

Consistent with DBA-ES §3.4, the CntlCo function may simultaneously carry a functional capability Design Basis (this document) and an AI Governance Design Basis under the AI Governance series. The dual-DB interaction at the CntlCo is more immediate than at any other DBA-ES instrument because the CntlCo's SL layer is already deeply AI-dependent: state estimators, contingency analysis tools, and AGC optimization all incorporate ML components in modern EMS implementations.

The Bright Line for CntlCo AI tools is established in the acceptance criteria (Section 8) and operates as follows. ML systems — state estimator algorithms, contingency analysis heuristics, AGC optimization — are deterministic, formally testable, and operate below the Bright Line with autonomous operation permitted. Gen AI systems — situational awareness synthesis tools, emergency procedure guidance generators, predictive dispatch recommendation systems — are non-deterministic and require a human Command Broker above the Bright Line. The licensed system operator on duty is the natural Command Broker for CntlCo Gen AI tools: NERC certification establishes the qualification standard, and the operator's legal accountability under NERC's rules of procedure provides the governance framework.

The AI Governance Design Basis for the CntlCo must address the adversarial ML attack on the SL layer (SL-1 adversarial bad data injection) as an AI governance concern as well as a physical design basis concern. If the state estimator's ML components are susceptible to adversarial inputs that cause systematic divergence without triggering bad data flags, the ML system has been effectively compromised as a Command Broker input. The AI Governance Design Basis must establish robustness requirements for CntlCo ML systems against adversarial input manipulation that go beyond the random measurement error tolerance addressed in standard EMS bad data detection.

Section 10 — Open Items and Deferred Items

Item	Description	Resolution Criterion
DI-1	DBA Consequence Envelope (v0.2). The full DBA non-adversarial consequence envelope is deferred to v0.2. This includes: bounding event identification for each postulated event category across all three layers; non-adversarial consequence analysis at Tier 1 (CntlCo functional degradation), Tier 2 (bulk electric system reliability impact), and Tier 3 (critical service curtailment); BCC transition time calibration against cascade initiation timeline; and quantitative minimum operator complement specification.	DBA consequence envelope completed in DBA-ES-CntlCo v0.2.
DI-2	MA Extension (v0.2). The military action extension is deferred to v0.2. This includes: MA extension applicability assessment; application of the four MA characteristics to each postulated event category; governing MA scenarios, including coordinated attack on multiple BA control centers within an RC area (XC-4); RC and TOP MA extension scope; and the adversarial cognitive overload strategy (deliberate simultaneous contingency generation to exceed operator cognitive capacity, HL-2).	MA extension completed in DBA-ES-CntlCo v0.2.
DI-3	RC authority chain communications design basis. The mechanism for maintaining RC-to-BA authority chain communications during primary control center loss (PL-1, PL-2) or during PL-4 communications infrastructure failure is not specified in this instrument. The RC extension scope in Section 5 identifies this as a requirement; the specific design basis for the authority chain communications is deferred.	RC authority chain communications design basis specified in v0.2 RC extension development.
DI-4	Manual operations mode minimum capability specification. The acceptance criterion for manual operations mode capability (Section 6) requires specification of the minimum design basis duration for manual operations and the minimum functional capabilities that must be maintained in manual mode. These specifications require calibration against the BA area's specific topology, load profile, and contingency risk profile and cannot be established at the series level.	Manual operations minimum capability specifications are established in the site-specific design basis implementation.

DI-5	SL-1 adversarial bad data injection design basis. The state estimator bad data detection acceptance criterion identifies adversarially manipulated SCADA data as a design basis concern. The specific design basis for bad data detection capability against systematic adversarial injection — as distinguished from random measurement errors — requires analysis of the adversary's data manipulation capability and the state estimator's statistical detection thresholds. This analysis is deferred.	Adversarial bad data injection design basis for the state estimator established in v0.2 SL analysis.
DI-6	Dual-DB AI Governance interaction points. Section 4.6 identifies Bright Line boundaries for CntlCo AI tools. The specific AI governance interaction points for CntlCo — the boundary between ML state estimation tools (below Bright Line) and Gen AI situational awareness synthesis tools (above Bright Line, Command Broker required) — require detailed analysis of the EMS AI architecture at each BA. This analysis is deferred to v0.2.	Bright Line boundary drawn for CntlCo EMS AI applications in v0.2.

Section 11 — References

Regulatory and Standards References

North American Electric Reliability Corporation. BAL-001-2: Real Power Balancing Control Performance. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. BAL-002-3: Disturbance Control Standard. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. EOP-011-2: Emergency Operations. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. IRO-001-5: Reliability Coordination — Responsibilities and Authorities. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-008-6: Incident Reporting and Response Planning. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

Government and Policy References

U.S. Department of Energy and North American Electric Reliability Corporation. "Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations." Washington, D.C.: DOE, April 2004.

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

Presidential Policy Directive 21 (PPD-21): Critical Infrastructure Security and Resilience. Washington, D.C.: The White House, February 2013.

Technical References

Kundur, Prabha. Power System Stability and Control. New York: McGraw-Hill, 1994. (Foundational reference for state estimation, AGC, and stability analysis in the EMS context.)

Wood, Allen J., Bruce F. Wollenberg, and Gerald B. Sheblé. Power Generation, Operation, and Control. 3rd ed. Hoboken: John Wiley & Sons, 2013. (State estimation and energy management system architecture.)

Abur, Ali and Antonio Gómez Expósito. Power Systems State Estimation: Theory and Implementation. New York: Marcel Dekker, 2004. (Bad data detection in state estimators.)

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES FC5 and CntlCo Gap Analysis. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Predecessor document.)

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-TX-FC3: HVDC Interconnect. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GC-FC4: Natural Gas Compressor Station. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-DI-FC5: Distribution Substation with Edge Generation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.